Mohammad Ammar | Aspiring AI/ML Researcher

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Education

Khoury College of Computer Sciences, Northeastern University
Master, Artificial Intelligence — *Sep 2025 – Dec 2027*

Zakir Husain College of Engineering and Technology, AMU
Bachelor, Computer Engineering — *Jul 2020 – Apr 2024*

Work

University of North Dakota
Research Intern — *Jun 2024 – May 2025*
- Developed an advanced Reinforcement Learning framework (RLDM-UAV) that integrates GPS spoofing detection and mitigation on Unmanned Aerial Vehicles (UAVs), achieving a **detection accuracy of over 96% and a task completion rate exceeding 91% under high-entropy conditions**.
- Implemented a Deep Q-Network (DQN) algorithm capable of real-time decisions in under 23 ms on ARM-based CPUs and proposed an **innovative experience replay mechanism that improved learning efficiency by 6x compared to baseline methods**.

University of Kiel
Research Intern — *Oct 2024 – Jan 2025*
- Developed a novel deep learning architecture, MicroCrackAttentionNeXt, for micro-crack detection in high-resolution spatio temporal images, achieving a **classification accuracy of 95.7% and an F1-score of 0.957** on imbalanced datasets.
- Leveraged attention mechanisms and multi-scale feature extraction to enhance model performance, outperforming traditional CNNs and transformer-based models in both accuracy and computational efficiency.
- Used Manifold Learning techniques to visualize high-dimensional feature spaces, providing insights into model decision-making processes.

Marine Technological Society - Autonomous Underwater Vehicle Club ZHCET — **https://www.auvzhcet.com/tech**
Chairperson — *May 2023 – Apr 2024*
- **Led 60 students** across mechanical, electronics, computer, and management teams, ensuring smooth administrative and technical operations.
- Enhanced the ROUV software stack by developing joystick control based on port-forwarding, using Python, which reduced system response time threefold. Implemented custom mapping functions for control input, improving thrust values for refined system operations.

Texas A&M University
Summer Research Intern — *Apr 2023 – Dec 2023*
- Conducted research on cybersecurity threats in maritime assets, analyzing their potential impact on human health and safety.
- Developed a comprehensive framework for assessing and mitigating cyber risks in maritime environments, contributing to the safety and security of maritime operations.

Marine Technological Society - Autonomous Underwater Vehicle Club ZHCET — **https://www.auvzhcet.com/tech**
Computer Team Member — *Jul 2022 – May 2023*
- Spearheaded the development of custom datasets for vision tasks such as buoy detection and docking station recognition.
- Utilized **Pytorch and ONNX** to quantize and deploy the Yolo Model on edge device, **gaining 9x speedups**.

Skills

AI & Machine Learning: Advanced **Keywords**: Python, PyTorch, Hugging Face, Ray RLlib, LangChain, TensorFlow, OpenCV, Pandas, Numpy, Matplotlib, TensorBoard, Pydantic, ChromaBD

Software & Tools: Advanced **Keywords**: ROS, Git, Linux, VS Code, LaTeX, Google Colab, Hydra, Poetry

Computer Science Fundamentals: Advanced **Keywords**: Data Structures, Algorithms, Object-Oriented Programming, Design Patterns, Software Development Life Cycle, DBMS, Networking, Operating Systems, Compiler Design, Parallel Computing

Publications

IEEE Internet of Things Journal
Reinforcement Learning Driven Integrated Detection and Mitigation of UAV GPS Spoofing Attacks

IEEE 2023 International Conference on Recent Advances in Electrical, Electronics & Digital Healthcare Technologies
Optical Coherence Tomography Image Classification using Light-weight Hybrid Transformers

MDPI Sensors Journal
MicroCrackAttentionNeXt

MusIML Workshop – NeurIPS 2025
From Rules to Pixels: A Decoupled Framework for Segmenting Human-Centric Rule Violations